

SSRN WORKING PAPER

A Designed-for-Test Measurement of Phantom-Brand Presence in Large Language Model Outputs

Pre-Registered Evidence from Bed Bath & Beyond, with Pier 1 as Structural Comparator

Working Paper · Version 0.7 · Designed-for-Test (Household Goods)

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4 May 2026

Working paper. Not under peer review. Pre-registered.

Abstract

This paper reports a designed-for-test measurement of brand visibility in large language model (LLM) outputs for Bed Bath & Beyond (BBB), thirty-seven months after the brand's Chapter 11 bankruptcy and online-only revival under post-acquisition ownership. Pier 1 — a smaller-footprint structural comparator with the same fate type but seventy months post-collapse — was added to the household-goods registry. Eight hypotheses with explicit numerical thresholds were committed to a pre-registration document locked on 4 May 2026 prior to any data collection. The measurement comprised six prompts × six models × eight runs (n = 288 successful measurements after a pre-registered failed-call recovery cycle) across model lineups from Anthropic, OpenAI, Google, and xAI. BBB's aggregate raw AI Presence was 38.2 percent, with a cross-model spread of 41.6 percentage points; Pier 1 surfaced at 0.0 percent across the same prompts and models. Six of eight pre-registered hypotheses were confirmed at the raw-rate level. Manual valence review of every BBB mention (88 percent strict inter-rater agreement on a stratified 25-row spot-check audit) reframes the result: the naive-phantom rate is 1.7 percent, and 88 percent of BBB mentions are surfaced with disclaimer indicating awareness of the entity change. The paper develops recommendation-slot persistence — the proposition that brand-mention pathways become reinforcement-encoded structures that persist beyond entity changes — as a candidate hypothesis adjacent to but structurally distinct from existing AI-Surfacing hypotheses. The reframe is exploratory; construct validity remains open and is the proposed object of a Phase 3 designed-for-confirmation study.

Keywords

AI brand visibility; phantom brands; recommendation-slot persistence; pre-registered measurement; designed-for-test methodology; brand-visibility measurement; large language models; AIAS

JEL Classification

M31 (Marketing) — primary; L86 (Information and Internet Services; Computer Software); L81 (Retail and Wholesale Trade; e-Commerce); G33 (Bankruptcy; Liquidation)

Paper status

Working paper. Not under peer review. Designed-for-test follow-up study to González Castro (2026), "AI Presence Measurement Across Consumer Categories" [SSRN ID forthcoming, v0.6]. Companion empirical artifact to González Castro (2026), "Tri-System Brand Growth: The AI Mediation Layer" [SSRN ID forthcoming]. Methodological reference: González Castro (2026), "AIAS Presence Measurement Protocol v1.1" [SSRN ID forthcoming]. Pre-registration document (PRE_REGISTRATION_household_v1.0.md, locked 4 May 2026) and underlying datasets released alongside this paper.

1. Introduction

In the cross-category baseline reported as v0.6 of the AIAS measurement program (González Castro 2026, "AI Presence Measurement Across Consumer Categories"), a single-category observation emerged that no other pattern in the dataset captured. Mint, the personal finance application Intuit decommissioned in March 2024, appeared in 44 percent of model responses to category prompts about personal finance applications 25 months after the brand had ceased to operate as a current consumer product. The pattern was termed phantom-brand surfacing. The single-category basis prevented the observation from being claimed as a structural property of LLM-mediated brand visibility versus an artefact of one category's discourse profile.

This paper reports a designed-for-test follow-up to that observation. Bed Bath & Beyond (BBB), a household-goods retailer that filed for Chapter 11 bankruptcy in April 2023, liquidated its physical stores, and was acquired through subsequent restructuring by Overstock — which rebranded itself to Beyond, Inc. and relaunched bedbathandbeyond.com as an online-only entity — was selected as the test subject. The category (household-goods retail) and the brand-disruption type (collapse plus online-only revival under post-acquisition ownership) differ structurally from the v0.6 Mint case (single-application decommissioning, no revival). Replication of phantom-brand surfacing across two such cases would substantially strengthen the claim that the pattern is a structural property of LLM-mediated brand visibility rather than a one-category observation.

Pier 1 — a smaller-footprint household-goods retailer with the same broad fate type (collapse plus online-only revival) but a different combination of variables (smaller pre-collapse retail footprint; longer time-since-collapse, approximately 70 months) — was added to the household-goods registry as a structural comparator. Pier 1's inclusion enables a within-category test of whether pre-collapse footprint size and time-since-collapse function as separable drivers of phantom signal. The Mint v0.6 observation could not isolate these variables; the BBB plus Pier 1 design can.

Eight hypotheses with explicit numerical thresholds were committed to a pre-registration document (PRE_REGISTRATION_household_v1.0.md) locked on 4 May 2026 prior to any data collection (Nosek et al. 2018). The hypotheses isolated specific candidate mechanisms: phantom rate (H1), cross-model variance (H2), within-lab freshness effects (H3, H4), cross-lab cutoff effects (H5), prompt-frame sensitivity (H6), structural comparator effect (H7), and rebrand propagation (H8). Pre-registered predictions and outcomes are reported as Table 1 in §4.

The headline empirical result confirms phantom replication. BBB's aggregate raw AI Presence was 38.2 percent across all 288 successful measurements, within four percentage points of Mint's 44 percent in v0.6. The H1 pre-registered threshold of 25 percent was cleared by 13 percentage points. Pier 1 surfaced at exactly 0.0 percent across all 288 measurements (strongest possible form of H7).

A second-stage manual valence review of every BBB mention reframes the result. Of 110 BBB mentions across 288 measurements, only 1.7 percent of measurements presented BBB as a current live retailer with no caveat (the naive-phantom rate). In 88 percent of BBB mentions, the model demonstrated knowledge that the entity had changed — surfacing the brand alongside acknowledgment of the closure, the rebrand,

or the online-only status. The phantom is not a knowledge gap; it is a recommendation slot that the model fills with disclosure rather than vacates.

This observation motivates the theoretical reframe developed in §5: recommendation-slot persistence, the proposition that brand-mention pathways in LLM outputs are reinforcement-encoded structures that persist beyond entity status changes, adjacent to but structurally distinct from existing AI-Surfacing hypotheses (Default Reinforcement; discourse-language bias; freshness-lag). The reframe is exploratory. Construct validity — whether consumers in disrupted categories act on caveated recommendations of defunct brands — remains open and is the proposed object of a Phase 3 designed-for-confirmation study.

2. Method

The measurement followed the AIAS Presence Measurement Protocol v1.0 (locked before v0.6 cross-category findings were published and used unchanged for this measurement; superseded for forward measurements by v1.1, González Castro 2026).

2.1 Designed-for-Test Selection and Brand Registry

The category was selected explicitly as a designed-for-test target for Pattern 6 (phantom-brand surfacing) from v0.6. BBB at approximately 37 months pre-measurement is the cleanest available test of whether the v0.6 Mint observation generalizes to a different category and disruption type.

The household-goods brand registry comprises 21 retailers spanning four tiers. Seven incumbents (Target, Amazon, Walmart, Costco, IKEA, HomeGoods, Wayfair); the Williams-Sonoma Inc. corporate-portfolio mid-tier (Williams-Sonoma, Pottery Barn, West Elm) plus specialty retailers (Crate & Barrel, World Market, Sur La Table, Container Store, At Home); four direct-to-consumer challengers (Brooklinen, Parachute, Boll & Branch, Quince); the phantom test subject (Bed Bath & Beyond); and the structural comparator (Pier 1, which collapsed in 2020). The category boundary excludes pure furniture retailers, online marketplaces, and grocery, preserving BBB's pre-collapse cognitive slot of "where consumers go specifically for home essentials."

2.2 Prompt Set

Six prompts were anchored to Category Entry Points (CEPs) per AIAS Protocol §3.1: FUNCTIONAL_WHY ("best store to buy household essentials"), CONTEXTUAL_WHEN ("wedding registry retailers"), CONSTRAINT_WITH ("affordable furnishing on a tight budget"), IDENTITY_HOW_FEELING ("stores people actually shop at when setting up a household"), DISCOVERY ("emerging or innovative home goods brands"), and COMPARISON ("compare leading household goods retailers"). Neither BBB nor Pier 1 was named in any prompt. Naming the test subject in the prompt was a v0.6 design issue (the v0.6 personal-finance discovery prompt named Mint by name) that compromised that specific prompt's phantom signal; the v0.7 design corrects this.

2.3 Model Lineup

Six frontier models from four labs spanned the production lineup as of May 2026: Anthropic Claude Sonnet 4.6 and Anthropic Claude Opus 4.7 (within-Anthropic generational comparison), OpenAI gpt-5.4-mini and OpenAI gpt-5.5 (within-OpenAI mini-vs-flagship comparison), Google Gemini 2.5 Flash, and xAI Grok 4.1 Fast. Grok 4.1 Fast carries the oldest knowledge cutoff in the lineup (November 2024), which spans the active 2023–2024 BBB bankruptcy and rebrand discourse. Within-lab pairs enable a clean test of freshness and reasoning effects with training-pipeline held approximately constant.

2.4 Measurement Volume and Failed-Call Recovery

Each prompt × model combination ran eight times at temperature 0.7 where supported, producing a target of 288 measurements. An initial 67 percent completion rate from API failures was restored to 100 percent through a pre-registered failed-call recovery cycle, yielding 288 successful measurements (six prompts × six models × eight runs).

2.5 Brand Mention Extraction

Brand mentions were extracted using gpt-5.4-mini at temperature 0 with structured-output classification, producing canonical brand mentions per response with rank, sentiment, and primary-recommendation flags. Mentions of brands not in the registry were retained on an unknown-mentions list and used as a continuing diagnostic of registry completeness (see §6 Limitations on TJ Maxx and Marshalls).

2.6 Manual Valence Review and Spot-Check Audit

A second-stage manual review classified every BBB mention along two axes. The valence taxonomy comprised five levels: live_recommendation (BBB presented as fully live with no caveat); live_with_caveat (BBB surfaced with disclosure of the closure or rebrand); status_correction (model explicitly states BBB is closed and does not recommend it); historical_reference (BBB mentioned only in past tense as a former retailer); and ambiguous. The entity-reference taxonomy comprised four levels: legacy_brand (Bed Bath & Beyond name only); corporate_parent (Beyond, Inc. name only); both (both names referenced); unclear.

A total of 110 BBB mentions were classified. An independent spot-check audit was conducted on a stratified 25-row sample drawn across the valence categories, producing 88 percent strict agreement with the gpt-5.4-mini-derived classifier. Three boundary disagreements clustered on the caveated/correction/historical adjacency. Zero disagreements occurred on the headline naive-phantom labels: all five live_recommendation rows verified correct on independent review. The classifier-vs-audit disagreement budget therefore does not affect the headline 1.7 percent naive-phantom rate but introduces a small uncertainty on aggregate aware-mode counts (treated in §6).

2.7 Pre-Registration

Eight hypotheses with explicit numerical thresholds were locked in PRE_REGISTRATION_household_v1.0.md on 2026-05-04, prior to any data collection. Each hypothesis

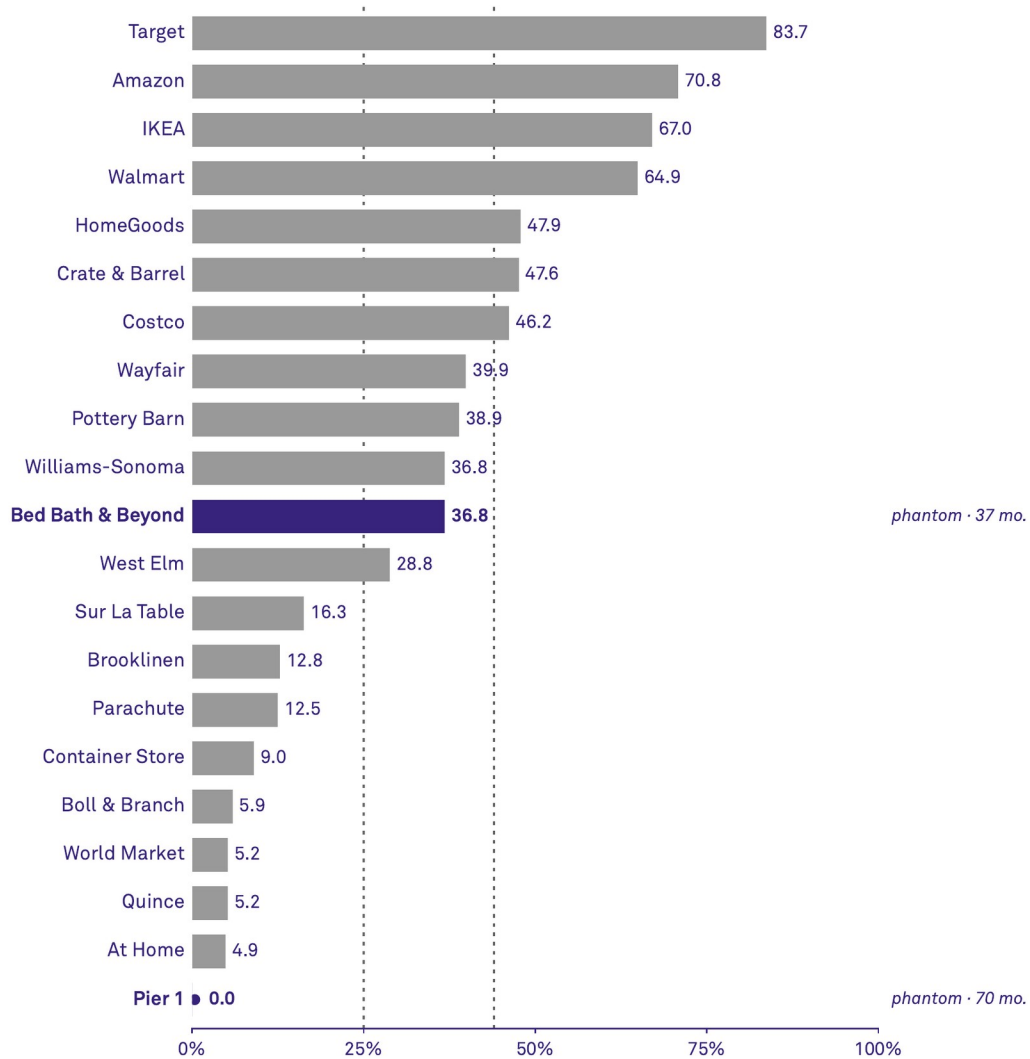
isolated a specific candidate mechanism. Hypotheses, predicted thresholds, and outcomes are reported in Table 1 (§4).

3. Results

AI Presence — Household Goods Retail

21 brands ordered by Presence. Bed Bath & Beyond and Pier 1 highlighted as phantom and comparator.

H1 threshold Mint v0.6 (44%)



Source: Third System AI Presence Index v0.7 · n=288 measurements (6 prompts × 6 models × 8 runs) · 4 May 2026

Figure A. Twenty-one household-goods retailers by AI Presence. Bed Bath & Beyond at #11 (36.8%) sits between Williams-Sonoma and West Elm; Pier 1 at #21 (0.0%) appears in zero of 288 measurements. The phantom signal is structurally specific.

3.1 Finding 1 — Phantom-brand surfacing replicates

Note on aggregate Presence values. Two slightly different BBB Presence values appear in this paper. Figure A reports 36.8% (structured-output extractor across the 21 retailers in the registry); the 38.2% figure used throughout this section and §3.2–§3.5 derives from the manual valence review (110 BBB mentions across 288 measurements). The 1.4-point gap reflects within-response repeated mentions and edge cases that the manual review captured but the structured extractor missed. Both values are preserved in the OSF deposit; the manifest documents the reconciliation.

BBB's aggregate raw AI Presence across all 288 measurements was 38.2 percent — within four percentage points of Mint's 44 percent in v0.6 personal finance. The pre-registered H1 threshold of 25 percent was cleared by 13 percentage points in a category where the brand has not operated as a physical retailer for 37 months.

The result holds across model providers. Of the six models in the lineup, all six surfaced BBB at meaningfully non-zero rates: Anthropic Opus 4.7 at 56.2 percent, Anthropic Sonnet 4.6 at 41.7 percent, OpenAI gpt-5.4-mini at 50.0 percent, xAI Grok 4.1 Fast at 50.0 percent. The two lowest — Google Gemini 2.5 Flash at 16.7 percent and OpenAI gpt-5.5 at 14.6 percent — surfaced BBB at rates substantially higher than the H1 floor.

The household-goods leaderboard places BBB at #11 of 21 registered retailers, between Williams-Sonoma (36.8 percent) and West Elm (28.8 percent). BBB surfaces at rates above 13 of 21 registered live retailers including Brooklinen, Parachute, Container Store, Boll & Branch, Quince, and five others currently operating in the category.

Two phantoms, two different fates

Same fate (collapse → online-only revival), different pre-collapse footprints — 38.2 pp gap.

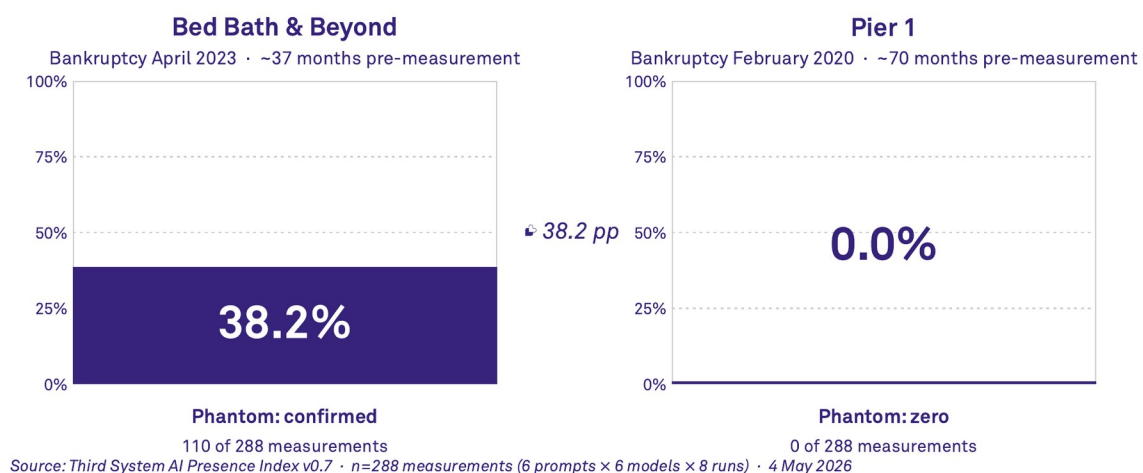


Figure 1. BBB and Pier 1 share a fate type (collapse plus online-only revival) but differ on years-since-collapse and pre-collapse footprint. BBB at 38.2 percent phantom; Pier 1 at exactly 0.0 percent across all 288 measurements.

Pier 1 — added explicitly to test whether pre-collapse footprint size and time-since-collapse are separable variables — surfaced in zero of 288 measurements: across every model, every prompt, and every run. The

phantom hypothesis is sharpened by this contrast: BBB at 38.2 percent and Pier 1 at 0.0 percent in the same registry, against the same prompts, across the same models. Two retailers with structurally similar fates but different pre-collapse footprints and different times-since-collapse produced dramatically different phantom signals.

A single measurement cannot fully separate the two variables; both contribute. The result establishes that a brand's pre-collapse cumulative editorial weight matters for phantom persistence, and that a smaller pre-collapse footprint combined with longer time-since-collapse is consistent with a phantom signal that drops to zero. H7 was confirmed in its strongest possible form. Cross-model variance for BBB (41.6-percentage-point spread between Opus 4.7 at 56.2 percent and gpt-5.5 at 14.6 percent) confirmed H2 (≥ 25 -point threshold).

3.2 Finding 2 — The phantom is a recommendation slot, not a knowledge gap

Of 110 BBB mentions across 288 measurements, the valence breakdown is: 5 live_recommendation (4.5 percent of BBB mentions, 1.7 percent of all measurements — the model presents BBB as fully live with no caveat); 72 live_with_caveat (65.5 percent of BBB mentions, 25.0 percent of all measurements — the model surfaces BBB with disclosure of the closure or rebrand); 7 status_correction (6.4 percent of BBB mentions, 2.4 percent of measurements — the model explicitly states BBB is closed and does not recommend it); and 25 historical_reference (22.7 percent of BBB mentions, 8.7 percent of measurements — the model mentions BBB only in past tense as a former retailer).

The headline number that drove the v0.6 Mint finding — and that confirmed H1 in this measurement — is the raw mention rate, ignoring valence. At the valence level, the picture inverts. The model is not blindly recommending BBB as a live retailer. The model mostly surfaces BBB with explicit disclosure of the closure or rebrand. In 88 percent of BBB mentions, the model demonstrates knowledge that the entity has changed. The naive-phantom rate — model presenting BBB as fully live with no caveat — is 1.7 percent of all measurements. Yet the brand still surfaces in 38.2 percent of responses. The gap between the model's correct knowledge of the entity's status and the model's continued surfacing of the brand is where the structural finding sits.

How each model handles Bed Bath & Beyond

Per-model valence breakdown of BBB mentions (denominator: 48 measurements per model).

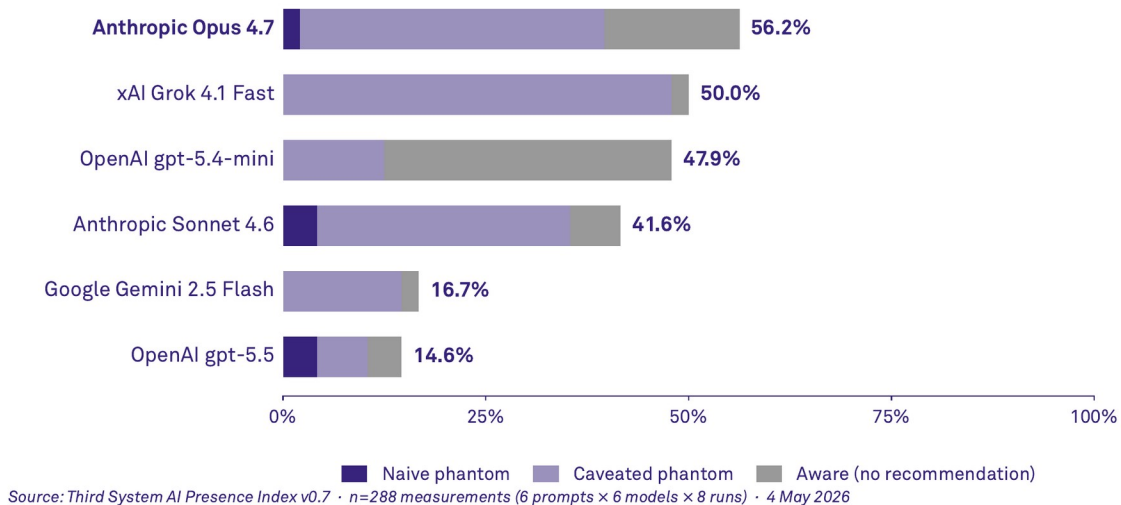


Figure 2. Per-model valence breakdown of BBB mentions (denominator: 48 measurements per model). Naive phantom is 1.7 percent of all measurements; caveated phantom is 25.0 percent. The model demonstrates knowledge that the entity has changed in 88 percent of mentions and surfaces the brand anyway, with disclosure.

A theoretical reframe is proposed: phantom-as-recommendation-slot, not phantom-as-knowledge-gap. The phantom in LLM mediation, on this account, is not a lag in what the model knows. The phantom is in what the model's recommendation set is structured to fill. Bed Bath & Beyond's accumulated editorial volume in AI training corpora — decades of comparison articles, registry guides, household-essentials listicles, store-by-store rankings — encoded a recommendation slot. The slot persists in the model's response pathway regardless of the entity's current operating status. When the entity changes, the model does not vacate the slot; the model fills the slot with disclaimer.

The construct-validity implication is that AI-mediated visibility outlasts entity integrity in disrupted categories. Brands that have collapsed, rebranded, or merged continue to occupy recommendation slots that consumers querying the model encounter, with disclosure. This represents a stronger structural claim than the naive lag interpretation offered for Mint in v0.6. Naive lag suggests training-data freshness as the variable to optimize. Recommendation-slot persistence suggests brand-mention pathways are reinforcement-encoded structures that retraining does not directly address. The same brand can occupy different recommendation slots across different models, and across different prompts within the same model, even when every model's underlying knowledge of the entity is identical (see §3.4 below).

3.3 Finding 3 — Within-lab and cross-lab patterns in phantom handling

The pre-registered H3 prediction — Anthropic Sonnet 4.6 BBB Presence > Anthropic Opus 4.7 BBB Presence by ≥ 15 points — was disconfirmed at the raw-Presence level. Opus surfaced BBB more often than Sonnet (56.2 percent vs. 41.7 percent), reversing the predicted direction by 14.5 points. The valence breakdown supports the underlying hypothesis at a different level. Of Opus's 27 BBB mentions, 96.3 percent were handled responsibly (caveated phantom plus aware mode). Of Sonnet's 20 BBB mentions,

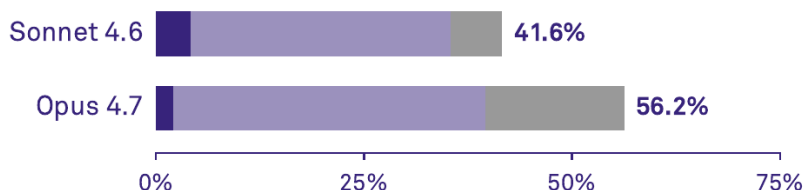
90.0 percent were handled responsibly. Opus surfaces BBB more often, but a higher fraction of its mentions cluster in the aware-mode category (status correction or historical reference) rather than recommendation.

Within-lab freshness

Newer = more responsible, not less frequent

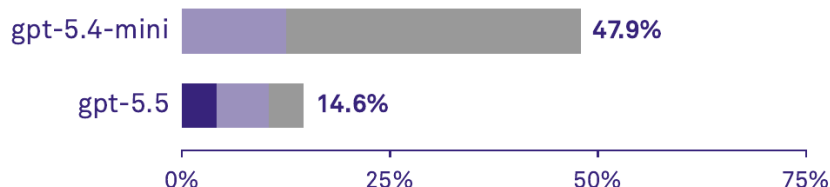
Anthropic

Sonnet 4.6 → Opus 4.7



OpenAI

mini → 5.5 (newer)



Source: AIPI v0.7 · 4 May 2026

Figure 3. Within-lab freshness comparison. Newer or larger models do not reduce raw phantom rate; they shift the per-mention valence distribution toward caveated or aware handling.

The pattern requires a careful claim. Newer models do not necessarily phantom-mention less. They phantom-mention with different valence handling. The freshness/reasoning differential between Sonnet 4.6 and Opus 4.7 appeared at the handling level — Opus is more likely to disclose the closure or historicize the mention — not at the frequency level. The recommendation slot persists; the disclosure improves.

The within-OpenAI comparison (gpt-5.4-mini vs gpt-5.5; H4 reported descriptively in the pre-registration) shows the same structure. gpt-5.5 surfaced BBB at the lowest rate of any model in the lineup (14.6 percent). The freshness-decay hypothesis would predict this is the cleanest model. Of those mentions, 4.2 percent were naive-phantom and 6.2 percent were caveated. By contrast gpt-5.4-mini surfaced BBB at 50.0 percent, with 35.4 percent in aware mode — the highest aware rate of any cell in the dataset. gpt-5.4-mini, the older and smaller OpenAI model, handled BBB more correctly per-mention than the newer flagship.

The cross-lab comparison confirmed H5: Grok 4.1 Fast — with the oldest knowledge cutoff in the lineup (November 2024) — produced phantom Presence at 50.0 percent, edging Sonnet 4.6 at 41.7 percent by 8.3 points. Grok's pattern is its own structural type: caveated-phantom rate of 47.9 percent (the highest of any model); aware rate of 2.1 percent (among the lowest). Grok almost always recommends BBB and almost always with caveat. Of Grok's BBB mentions, 58.3 percent explicitly reference both entity names —

far higher than any other model. The candidate mechanism: Grok's older training cutoff captures the active 2023–2024 discourse around BBB's bankruptcy, the rebrand to Beyond, Inc., and the Overstock acquisition, locking in a "BBB-as-active-with-disclosure" pattern that newer models with later training data treat as more historicized.

The observation suggests each lab's training pipeline produces a distinguishable phantom signature — a structural property of the lab and its training/instruction-tuning practices, not just of the model's generation. Anthropic models surface BBB at high rates with mostly responsible handling. Grok surfaces BBB at high rates with always-disclosed handling. OpenAI mini is the most aware model, handling BBB as past-tense in a third of mentions. OpenAI flagship and Gemini Flash mention BBB rarely but inconsistently when they do.

3.4 Finding 4 — Prompt frame activates a temporal cognitive geometry

Per-CEP analysis reveals an outcome outside the pre-registration: phantom valence varies systematically with the cognitive frame the prompt activates. The same brand, on the same models, with six different prompts occupied different temporal slots depending on what the prompt asked.

In FUNCTIONAL_WHY prompts ("best store for bedding, towels, kitchen basics"), BBB raw Presence was 70.8 percent, with 60.4 percent of measurements producing caveated-phantom recommendations — the model surfacing BBB-as-still-relevant-online-retailer alongside disclosure of the physical-store closure. Functional queries elicited BBB-as-current. In IDENTITY_HOW_FEELING prompts ("stores people actually shop at when setting up a household"), BBB raw Presence was 81.2 percent — the highest of any prompt — but 43.8 percent of measurements produced aware-mode mentions (historical references or corrections). Identity queries elicited BBB-as-cultural-memory.

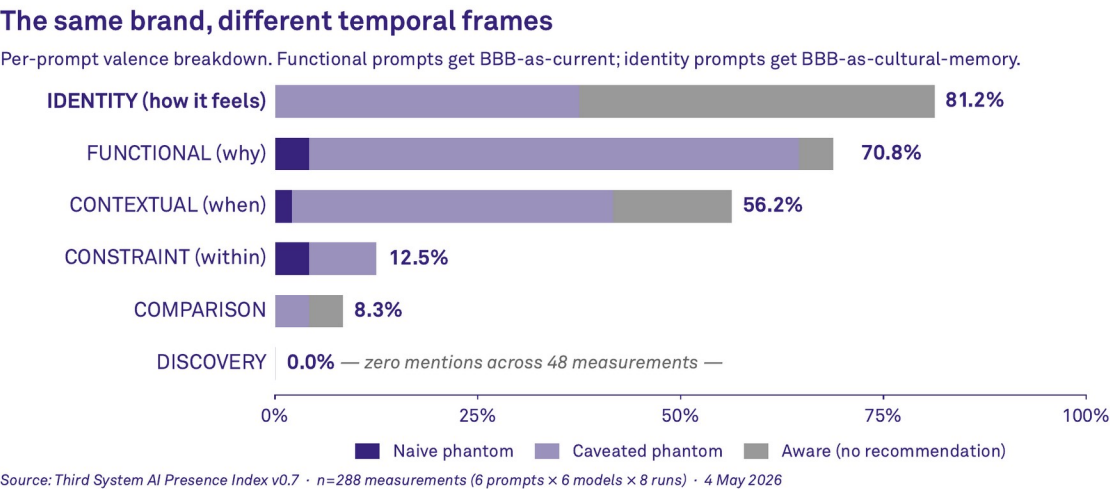


Figure 4. Per-CEP temporal frame breakdown. Functional prompts produce BBB-as-current; identity prompts produce BBB-as-cultural-memory; discovery prompts produce zero BBB mentions.

This temporal cognitive geometry was not anticipated in the v0.6 framework. BBB lives in different prompt frames as different temporal entities. When the model is asked functionally, BBB surfaces as a current

online retailer with disclosure. When the model is asked about cultural identity around home setup, BBB surfaces as a former retailer in past tense. Same brand, different prompts, different cognitive-time slots.

The construct-validity implication is that AI-mediated visibility is not single-valued, even within a category. The same brand can simultaneously occupy a current-recommendation slot in one prompt frame and a historical-anchor slot in another. Consumers querying different frames will encounter materially different brand presentations. For disrupted brands, the geometry produces a particular pattern: continued recommendation in functional frames, continued historical anchoring in identity frames, and absence in comparison frames where live competitors saturate the surface.

H6 was partially confirmed. Lowest activation was in DISCOVERY (0.0 percent, exactly as predicted) but highest activation was in IDENTITY (81.2 percent) and FUNCTIONAL (70.8 percent), not in CONTEXTUAL or COMPARISON as pre-registered. The COMPARISON-frame absence is explained by Risk #2 in the pre-registration design document: Walmart, Amazon, Target, and IKEA saturate the comparison frame at near-100 percent each, leaving structurally constrained headroom for BBB to surface in that prompt regardless of phantom strength.

3.5 Finding 5 — The rebrand has not propagated; the legacy brand has

H8 was confirmed. Of 110 BBB mentions, 0 percent referenced Beyond, Inc. as a corporate parent in isolation, 23.6 percent referenced both names with awareness they are related, and 75.5 percent referenced only the legacy "Bed Bath & Beyond" string. By either count — strict (Beyond, Inc. only) or inclusive (Beyond, Inc. mentioned at all) — the Beyond, Inc. share is below the 25-percent pre-registration threshold.

The cross-tabulation with valence reveals the structure. Of the 5 naive-phantom mentions (live_recommendation), 100 percent referenced only the legacy name. The model in naive-recommendation mode has not internalized the rebrand. Of the 72 caveated-phantom mentions, 65.3 percent reference only the legacy name and 34.7 percent reference both — caveated mentions are where the rebrand information lives, used to construct the caveat itself.

Where the rebrand lives in AI's mentions of BBB

Of 110 BBB mentions, the rebrand information shows up almost only in caveats. Naive recommendations stay pure-legacy.

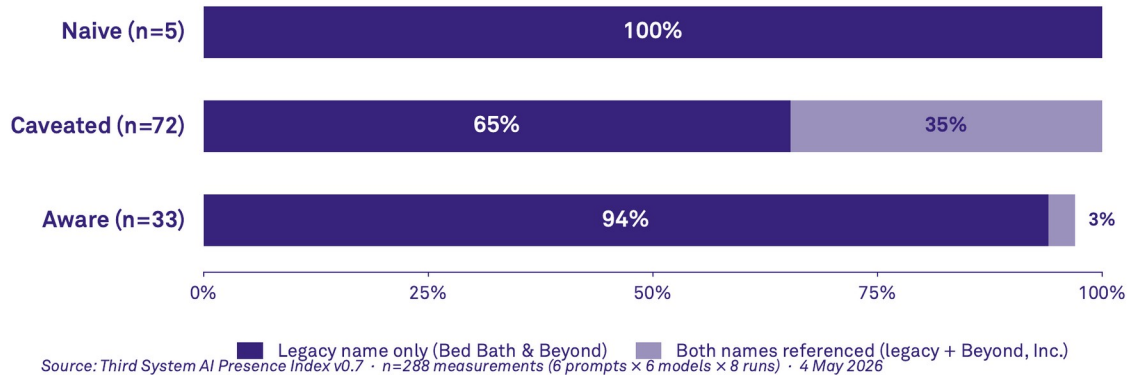


Figure 5. Rebrand reference distribution by valence type. Naive recommendations are 100 percent legacy-name-only. The rebrand information lives almost entirely in caveated mentions — deployed as part of the disclosure ("now operates as Beyond, Inc.") rather than internalized into the recommendation surface.

When the model knows about the rebrand, the rebrand information is deployed as part of the disclosure — for example, "Bed Bath & Beyond now operates as Beyond, Inc." — not as a replacement of the legacy name in the model's recommendation set. The recommendation set remains anchored on Bed Bath & Beyond. Beyond, Inc. functions as a footnote.

For brand-strategy interpretation, this implies that rebranding a disrupted entity into a new corporate identity does not propagate into the model's recommendation surface at the same rate that the legacy brand persists. The new entity name occupies a footnote slot, not a recommendation slot. The finding is specific to a particular rebrand pattern (distressed parent acquired, online-only revival under legacy domain) and does not necessarily generalize to other rebrand types — organic name changes for live brands, acquisition rebrands without distress, holding-company consolidation. A Phase 3 follow-up will measure additional rebrand patterns to test how broadly this asymmetry holds.

4. Pre-Registration Outcomes

Eight hypotheses with explicit numerical thresholds were locked in PRE_REGISTRATION_household_v1.0.md on 2026-05-04, prior to any data collection. Five were confirmed; one was partially confirmed; one was disconfirmed at the level it was pre-registered but supported at a different level (handling rather than frequency); one was reported descriptively. Predictions and outcomes are reported in Table 1.

Table 1. Pre-registered hypotheses, predictions, and outcomes.

H	Pre-registered prediction	Result	Status
H1	BBB Presence ≥ 25%	38.2% raw · 26.7% naive+caveated · 1.7%	Confirmed

H	Pre-registered prediction	Result	Status
		naive	
H2	Cross-model spread ≥ 25 pts	41.6 pts (Opus 56.2 $\rightarrow$ flagship 14.6)	Confirmed
H3	Sonnet > Opus by ≥ 15 pts	Opus > Sonnet by 14.5 pts (raw); Opus more aware per-mention	Disconfirmed at raw, supported at handling
H4	Within-OpenAI freshness (descriptive only)	mini > flagship by 35.4 pts (raw); mini more aware per-mention	Descriptive: opposite directions raw vs handling
H5	Grok $\geq$ Sonnet on phantom rate	Grok 50.0 > Sonnet 41.7 by 8.3 pts	Confirmed
H6	Highest in p2/p6, lowest in p5	Highest in p4 (81.2)/p1 (70.8); lowest in p5 (0.0)	Partially confirmed
H7	Pier 1 < BBB AND Pier 1 < 15%	Pier 1 = 0.0% across all 288 measurements	Confirmed (strongest form)
H8	Beyond, Inc. < 25% of BBB+Beyond	0% strict · 23.6% inclusive	Confirmed

Each hypothesis isolated a specific candidate mechanism. H1 tested whether the v0.6 Mint phantom observation generalizes to a different category, with the 25-percent threshold equal to roughly half the v0.6 signal — sufficient evidence that phantom-brand surfacing is a structural property of LLM mediation rather than a one-category artefact. H2 tested whether phantom signal varies meaningfully by model lineage; a wide spread suggests training-pipeline differences across labs and models. H3 tested the freshness hypothesis directly within Anthropic, with training pipeline approximately constant. H4 reported the within-OpenAI mini-vs-flagship comparison descriptively because the two models differ on both size and cutoff. H5 tested cross-lab cutoff effects through Grok's November 2024 training cutoff. H6 tested prompt-frame sensitivity. H7 — the structural comparator — tested whether pre-collapse footprint and time-since-collapse are separable variables. H8 tested rebrand propagation: whether the model's recommendation set has internalized the rebrand to Beyond, Inc.

5. Discussion: Recommendation-Slot Persistence

The v0.6 phantom-brand observation was framed as a property of training-data freshness: backward-looking corpora capture historical discourse, and a brand decommissioned after the corpus is collected continues to surface in responses for an indeterminate period. This framing predicts that the variable available for optimization is corpus freshness — that retraining or updating the model on more recent data will, over time, reduce phantom-mention rates.

The v0.7 result is inconsistent with that prediction at the mechanism level. In 88 percent of BBB mentions, the model demonstrates knowledge that the entity has changed. The model's underlying representation

of BBB is not stale. Yet the brand still surfaces in 38.2 percent of responses — frequently at saturation in functional and identity prompts, with structured disclosure of the closure or rebrand. The model knows the entity status. The brand surfaces anyway. Freshness is not the binding constraint.

A candidate alternative framing — recommendation-slot persistence — is proposed. On this account, brand-mention pathways in LLM outputs function as reinforcement-encoded structures within the model's response-construction logic. Bed Bath & Beyond's decades of accumulated editorial volume — comparison articles, registry guides, household-essentials listicles, store-by-store rankings — encoded a recommendation slot for the household-goods category. The slot is a property of the model's response pathway, not a property of the model's factual knowledge of the entity. When the entity changes, the model does not vacate the slot. The model fills the slot with disclosure language, drawing on its updated factual knowledge to construct the caveat.

This framing is offered as a candidate hypothesis adjacent to but structurally distinct from existing AI-Surfacing hypotheses (Default Reinforcement; discourse-language bias; freshness-lag). It is consistent with three observations from this measurement that the freshness-lag framing alone does not predict:

(a) Within-lab generational comparisons (Sonnet 4.6 vs. Opus 4.7; gpt-5.4-mini vs. gpt-5.5) show that newer models phantom-mention with different valence distributions, not at lower frequencies. The recommendation slot persists across model generations within a lab; only the disclosure language changes.

(b) Per-CEP variation (BBB at 81.2 percent in IDENTITY, 70.8 percent in FUNCTIONAL, 0.0 percent in DISCOVERY) shows that phantom signal depends on the cognitive frame the prompt activates. The slot is filled when the prompt activates a frame the brand was strongly editorially associated with (registry, household setup, kitchen basics) and is empty when the prompt activates a frame the brand never occupied (emerging-brands discovery).

(c) Rebrand non-propagation (Beyond, Inc. at 0 percent strict, 23.6 percent inclusive) shows that updated factual information about the entity does not displace the legacy brand in the model's recommendation set; it occupies a footnote slot used to construct the caveat.

The reframe is exploratory. It is offered as a candidate hypothesis to be tested in subsequent designed-for-test studies, including the longitudinal designs proposed in §7. The reframe predicts that retraining alone will not vacate phantom slots; it will redistribute the disclosure language used to fill them. A controlled test would measure the same disrupted brand on the same model across a major model update, comparing phantom Presence and valence distribution before and after. This is a direct test of the freshness-lag hypothesis the reframe is now contesting.

Construct validity remains open. The phantom finding is a property of LLM mediation, not directly of consumer behavior. Whether consumers in disrupted categories act on caveated recommendations of defunct or rebranded entities — and at what rates such recommendations affect consideration, purchase intent, or category share — is the separate question the AIAS Phase 3 construct-validity program will address. Both possible outcomes are publishable: a strong correlation establishes AI Presence as a leading indicator of consumer behavior in disrupted categories; a weak correlation establishes the index as a

measurement of LLM behavior with implications limited to channel optimization for AI-mediated visibility. Neither is established here.

6. Limitations

Six caveats apply to the findings reported here.

Single category. Phase 2 measures one designed-for-test category. Together with the v0.6 Mint observation, the BBB result moves the program from one-category claim to two-category claim — substantively defensible, not yet generalizable. Replication across additional category-disruption types is required before the phantom-as-recommendation-slot framing can be claimed as a general property of LLM-mediated brand visibility.

Inter-rater agreement on boundary cases. Manual review classifications used a single AI classifier (gpt-5.4-mini at temperature 0). The independent spot-check audit on a stratified 25-row sample produced 88 percent strict agreement with three boundary disagreements clustering on the caveated/correction/historical adjacency. The disagreements do not affect the headline naive-phantom rate (1.7 percent; all five live_recommendation rows verified correct on independent review) but introduce a small uncertainty budget on aggregate aware-mode counts.

Walmart and Amazon ceiling compression. Risk #2 in the pre-registration design document predicted that COMPARISON and FUNCTIONAL prompts would be ceiling-compressed by Walmart and Amazon at near-100 percent each, leaving constrained headroom for specialty brands including BBB. The prediction held. BBB's COMPARISON-prompt rate of 8.3 percent partly reflects this constraint and not phantom strength alone.

Registry leakage. The unknown-mentions list includes TJ Maxx (132 mentions) and Marshalls (129 mentions) as the highest unknowns — both TJX sister brands of HomeGoods, which is in the registry. Phase 3 measurements should add the TJX sister set to the registry. Threshold (Target's house brand) appears at 42 mentions, representing component-mode leakage rather than retailer-mode mentions; this does not affect BBB analysis.

Single point in time. Each measurement reflects model behavior on 4 May 2026. The findings do not describe how AI brand visibility moves over time. The recommendation-slot persistence framing predicts that retraining alone will not vacate the slot; this prediction will be testable when the same models update.

Construct validity unproven. AI Presence measures a real and stable property of the LLM tier, but whether that property correlates with consumer consideration, purchase intent, or sales is unknown. The phantom finding is a property of LLM mediation, not of consumer behavior; whether consumers in disrupted categories actually act on caveated recommendations of defunct brands is the separate question the AIAS Phase 3 construct-validity program will address.

7. Future Research

The recommendation-slot persistence framing, if it generalizes, predicts that retraining will not vacate phantom slots — only redistribute the disclosure language used to fill them. A controlled longitudinal test would measure the same disrupted brand on the same model across a major model update, comparing phantom Presence and valence distribution before and after. This is a direct test of the freshness-lag hypothesis the reframe contests.

A construct-validity study correlating AI Presence and valence distribution against external consumer-tracking data is also priority. For BBB specifically, candidate validators include Google Trends, consumer-awareness surveys, and DTC brand-tracking instruments. The recommendation-slot persistence framing predicts that consumer awareness of BBB will be lower than AI Presence — the slot persists in the model even when consumer attention has moved on. If the prediction holds, AI Presence is a lagging indicator in disrupted categories. If it inverts, AI Presence is leading consumer behavior. Both findings are publishable; neither is established.

The within-lab handling pattern (newer or smaller models showing different valence distributions than newer-flagship models within the same lab) suggests that LLM handling of disrupted brands is not a single-variable function of training cutoff or model size, but interacts with reasoning-mode and instruction-tuning practices. A measurement isolating model-size-vs-cutoff (running the same brand against same-cutoff models of different sizes within a lab) would clarify the mechanism.

Additional rebrand patterns — organic name changes for live brands, acquisition rebrands without distress, holding-company consolidation — would extend the v0.7 H8 finding beyond the specific pattern measured here (distressed parent acquired, online-only revival under legacy domain). Candidate cases include the Twitter rebrand to X, the Facebook to Meta corporate rebrand, and several pharmaceutical brand transitions.

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Declarations

Conflict of Interest

The author serves as Director, Corporate Brand Creative and Governance at Samsung Electronics America. The research presented here is independent of Samsung Electronics America and does not constitute Samsung research. No Samsung Electronics America data, personnel, or commercial interests influenced the design, conduct, analysis, or reporting of this study. Samsung Electronics America did not review the manuscript prior to posting.

Funding

Self-funded. No external funding sources contributed to this research.

Data Availability

Underlying datasets, the locked pre-registration document (PRE_REGISTRATION_household_v1.0.md, lock date 2026-05-04), and a detailed reconstruction manifest are deposited in an Open Science Framework project at <https://osf.io/ec6wh/>. The deposit contains aggregate analysis outputs (manual review distributions, per-brand Presence, valence and entity cross-tabulations, household-goods leaderboard, and per-CEP×per-model BBB grid). Provenance note: the original measurement run produced row-level CSV outputs in a code-execution environment that was disposed before the data was archived; the deposited CSVs are reconstructed from the printed transcript outputs of the measurement session. Aggregate analyses are fully recoverable; row-level data (response text, per-row brand extraction, per-row manual-review entity labels for status_correction / historical_reference / ambiguous categories, and per-row spot-check audit detail) is not. The OSF manifest documents recovery status, gaps, and known value discrepancies, including the leaderboard-vs-manual-review reconciliation discussed at the start of §3.1.

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